

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

Course Code:	CSA6501	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 2 - Model Comparison Presentation (BERT vs GPT vs T5 vs LLaMA)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
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Section / Batch:	CSE-AI	Date:	

Instructions to Students

- Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.
- Compare all four models: BERT, GPT, T5, and LLaMA.
- Explain the **architecture and working principle** of each model.
- Compare their **pre-training objectives and capabilities**.
- Map each model to appropriate **real-world applications**.
- Identify the **strengths and limitations** of each model.
- Support the presentation using **credible research papers, official documentation, benchmarks, or other reliable sources**.
- Include at least one **comparison table or diagram**.
- Use relevant **real-world case studies or application examples**.
- Conclude by identifying the **most suitable model for the selected application** and justify the decision.
- Include proper **references/citations** in the presentation.

Code of Conduct

I A.Sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A.Sai Rohit

SECTION A : MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

18. Compare the four models for information extraction, including named entity recognition and relation extraction. Explain the advantages and limitations of each.

Introduction

Information Extraction (IE) is the process of automatically extracting useful and structured information from unstructured text. Two important IE tasks are Named Entity Recognition (NER) and Relation Extraction (RE).

- Named Entity Recognition (NER): Identifies entities such as people, organizations, locations, dates and products.
- Relation Extraction (RE): Identifies relationships between entities, such as “*Elon Musk* → *founded* → *SpaceX*.”

BERT, GPT, T5 and LLaMA are all Transformer-based models, but they are designed differently. BERT is mainly focused on understanding text, while GPT and LLaMA are primarily generative models. T5 uses an encoder-decoder architecture and treats NLP tasks as text-to-text problems. Therefore, their suitability for information extraction differs.

1. BERT Architecture

BERT stands for Bidirectional Encoder Representations from Transformers. It uses a Transformer encoder-only architecture.

BERT reads the input from both left and right contexts simultaneously. This bidirectional understanding helps it understand the meaning of a word based on the complete sentence.

Working Principle

For NER, BERT processes a sentence and generates a contextual representation for every token. A classification layer can then assign an entity label to each token.

Example:

Input:

"Microsoft was founded by Bill Gates."

NER Output:

- Microsoft → ORGANIZATION
- Bill Gates → PERSON

For relation extraction, BERT can be fine-tuned to identify relationships between two entities. Research has also demonstrated BERT-based approaches for relation extraction.

Advantages

1. Excellent contextual understanding.
2. Very effective for NER and classification-based tasks.
3. Can be fine-tuned using relatively small labeled datasets.
4. Good accuracy for domain-specific information extraction.
5. Bidirectional attention helps identify the correct meaning of entities.

Limitations

1. BERT is not naturally designed for free-form text generation.
2. Usually requires task-specific fine-tuning.
3. Processing very long documents can be challenging.
4. A separate classification architecture may be required for complex relation extraction.
5. Large BERT models can require significant computational resources.

Best Use

BERT is highly suitable for:

NER, entity classification, sentence classification and structured information extraction.

2. GPT Architecture

GPT is based on a Transformer decoder-only architecture. It processes text from left to right and predicts the next token.

GPT models are designed primarily for text generation, but their ability to understand instructions allows them to perform information extraction through prompting or fine-tuning.

Working Principle

A user can provide a document and ask GPT to identify entities and relationships.

Example:

Input:

"Apple CEO Tim Cook announced a new product in California."

Possible Output:

- Apple → ORGANIZATION
- Tim Cook → PERSON
- California → LOCATION
- Tim Cook → CEO of → Apple

GPT-3 research demonstrated strong few-shot capabilities, where tasks can be specified through natural-language instructions and examples without traditional task-specific fine-tuning.

Advantages

1. Can perform NER and relation extraction using natural-language prompts.
2. Requires little or no task-specific training in zero-shot or few-shot settings.
3. Can extract information into formats such as JSON or tables.
4. Strong general language understanding and generation capabilities.
5. Flexible for multiple IE tasks using the same model.

Limitations

1. May generate incorrect or hallucinated entities and relationships.
2. Output can be inconsistent without carefully designed prompts.
3. Large GPT models can be expensive to run.
4. Exact token-level extraction may be less predictable than a specialized BERT model.
5. Sensitive information may require additional privacy and security controls.

Best Use

GPT is suitable for:

Flexible information extraction, zero-shot/few-shot extraction, document analysis and extracting structured information from diverse text.

3. T5 Architecture

T5 stands for Text-to-Text Transfer Transformer. It uses an encoder-decoder Transformer architecture.

Unlike BERT, which mainly produces representations for understanding, T5 converts NLP tasks into a common text-to-text format.

Working Principle

The input and output are both represented as text.

For example:

Input:

"extract entities: Apple CEO Tim Cook visited California."

Output:

"Apple | ORGANIZATION

Tim Cook | PERSON

California | LOCATION"

For relation extraction:

Input:

"extract relations: Tim Cook is the CEO of Apple."

Output:

"Tim Cook | CEO_OF | Apple"

T5's text-to-text approach makes it possible to use one general framework for different NLP tasks. Its pre-training includes corrupted-token reconstruction and its framework supports multiple downstream tasks.

Advantages

1. Flexible text-to-text framework.
2. Can perform both NER and relation extraction.
3. Suitable for generating structured extraction results.
4. Can be fine-tuned for domain-specific information extraction.
5. The same architecture can be adapted to many NLP tasks.

Limitations

1. Generation can sometimes produce incorrect information.
2. Usually needs fine-tuning for high-quality domain-specific extraction.
3. More computationally expensive than smaller encoder-only models for simple classification tasks.
4. Generated outputs may require post-processing.
5. Exact token-level entity boundaries can be harder to guarantee.

Best Use

T5 is suitable for:

Text-to-text information extraction, relation extraction and applications requiring both understanding and generation.

4. LLaMA Architecture

LLaMA is a family of large language models developed by Meta. Modern LLaMA models use a decoder-only Transformer architecture. Meta describes Llama 3 as a standard decoder-only Transformer with improvements such as grouped-query attention.

LLaMA models are designed as general-purpose foundation models and can be adapted for many NLP tasks.

Working Principle

LLaMA predicts the next token and can use prompts to perform information extraction.

Example:

Input:

"Extract all people, organizations and locations from:
'Google CEO Sundar Pichai spoke at an event in California.'"

Output:

- Sundar Pichai → PERSON
- Google → ORGANIZATION
- California → LOCATION

For relation extraction:

- Sundar Pichai → CEO_OF → Google

LLaMA 3 was evaluated on real-world tasks including classification, extraction and summarization, demonstrating its ability to handle information extraction among other applications.

Advantages

1. Strong general language understanding.
2. Can perform extraction through prompting.
3. Open model ecosystem provides opportunities for customization and fine-tuning.
4. Suitable for large-scale document analysis.
5. Can perform multiple NLP tasks using one foundation model.
6. Modern LLaMA versions support advanced capabilities and tool-based applications.

Limitations

1. Large models require substantial computational resources.
2. Generative output can contain hallucinations.
3. Extraction results may not always be deterministic.
4. Prompt design can significantly affect results.
5. Smaller specialized models may be more efficient for simple NER tasks.

Best Use

LLaMA is suitable for:

Large-scale document analysis, flexible information extraction, domain-specific fine-tuning and applications requiring multiple NLP capabilities.

5. Comparison Table

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder-only	Decoder-only	Encoder-decoder	Decoder-only
Main Purpose	Language understanding	Text generation	Text-to-text NLP	General-purpose generation
NER	Excellent	Very Good	Very Good	Very Good

Relation Extraction	Excellent with fine-tuning	Very Good with prompting/fine-tuning	Very Good	Very Good
Context Understanding	Excellent	Excellent	Excellent	Excellent
Zero-shot Extraction	Limited compared with generative LLMs	Excellent	Good	Excellent
Structured Output	Good with task-specific head	Very Good with prompting	Very Good	Very Good
Fine-tuning Requirement	Usually required	Optional	Often useful	Optional/fine-tuning useful
Hallucination Risk	Low compared with generative models	Higher	Moderate	Higher
Computational Cost	Low–Medium	High	Medium–High	High
Best Strength	Precise extraction	Flexible extraction	Unified text-to-text tasks	General-purpose and customizable extraction

6. BERT vs GPT vs T5 vs LLaMA for NER

For Named Entity Recognition, BERT is generally the strongest choice when the goal is precise and consistent token-level classification.

For example:

Sentence:

"Amazon opened a new office in Chennai."

BERT can classify each relevant token as:

- Amazon → ORGANIZATION
- Chennai → LOCATION

GPT and LLaMA can also identify these entities through prompting, but their generative nature means the output format may vary. T5 can generate the entities as text and can be fine-tuned specifically for NER.

NER Ranking for a Specialized System

BERT → T5 → LLaMA/GPT

BERT is preferred when accuracy, consistency and efficient inference are more important than general-purpose generation.

7. BERT vs GPT vs T5 vs LLaMA for Relation Extraction

Relation Extraction requires identifying relationships between entities.

Example:

Sentence:

"Satya Nadella is the CEO of Microsoft."

Expected relation:

Satya Nadella → CEO_OF → Microsoft

BERT can be fine-tuned to classify the relationship between entity pairs. This makes it effective for structured relation extraction.

GPT and LLaMA can identify relationships directly from natural-language instructions and can return structured formats such as JSON.

T5 can treat relation extraction as a text-generation problem and generate the required relationship.

Relation Extraction Ranking

For a specialized and structured system:

BERT → T5 → LLaMA → GPT

For zero-shot or flexible extraction:

GPT/LLaMA → T5 → BERT

The exact ranking depends on the dataset, model version, prompt, fine-tuning method and evaluation metric.

8. Real-World Application Example

Consider a news intelligence system that processes thousands of articles every day.

Example article:

"Microsoft CEO Satya Nadella announced a new AI initiative in London."

The system needs to identify:

Entities

- Microsoft → ORGANIZATION
- Satya Nadella → PERSON
- London → LOCATION

Relation

- Satya Nadella → CEO_OF → Microsoft
- Microsoft → ANNOUNCED → AI initiative

Model Selection

BERT: Excellent when the organization needs highly accurate entity and relationship classification.

GPT: Useful when articles are diverse and extraction instructions change frequently.

T5: Useful when the organization wants a unified text-to-text extraction pipeline.

LLaMA: Useful when the organization wants an adaptable foundation model that can be customized and deployed within an open model ecosystem.

9. Advantages and Limitations Summary

Model	Main Advantages	Main Limitations
BERT	Excellent contextual understanding, precise NER, strong fine-tuning performance	Requires task-specific training, not primarily generative
GPT	Flexible, strong zero-shot/few-shot extraction, natural-language prompting	Hallucination, cost, output variability
T5	Unified text-to-text framework, flexible for NER and RE	Generation overhead, may require post-processing
LLaMA	Powerful general-purpose model, customizable, open model ecosystem	High compute requirements, hallucination and consistency issues

10. Overall Comparison

The four models have different strengths.

BERT is mainly an understanding model and is particularly effective for structured information extraction tasks such as NER and relation classification.

GPT is a powerful generative model that can perform information extraction using prompts and examples, making it highly flexible.

T5 provides a unified approach by converting information extraction into a text-to-text generation problem.

LLaMA provides strong general-purpose language capabilities and can be customized for information extraction through prompting and fine-tuning.

Therefore, there is no single model that is best for every information extraction scenario.

11. Conclusion

For information extraction, especially Named Entity Recognition and Relation Extraction, BERT is the most suitable choice when the main requirement is accurate, consistent and structured extraction. Its bidirectional encoder architecture provides strong contextual representations, and it can be fine-tuned specifically for NER and relation extraction. The original BERT research demonstrated its effectiveness across multiple language-understanding tasks, while later research has applied BERT-based methods to relation extraction.

However, GPT and LLaMA are better choices when flexible zero-shot or few-shot extraction is required, while T5 is a strong option when the complete information-extraction pipeline is naturally expressed as a text-to-text task.

Therefore:

Best for specialized NER and structured IE → BERT

Best for flexible zero-shot/few-shot IE → GPT

Best for unified text-to-text IE → T5

Best for customizable general-purpose IE → LLaMA

The final model should be selected based on the application's requirements, available computational resources, accuracy requirements, need for fine-tuning, and tolerance for generative errors.

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33. A news organization wants to automatically summarize thousands of news articles. Compare BERT, GPT, T5, and LLaMA and recommend an appropriate model.

A news organization may receive thousands of articles every day. Reading every article completely is time-consuming for journalists and readers. Automatic text summarization uses Natural Language Processing (NLP) models to generate a shorter version of an article while preserving its important information.

There are two major types of summarization:

- Extractive summarization: Selects the most important sentences from the original article.
- Abstractive summarization: Understands the article and generates a new summary using its own words.

BERT, GPT, T5 and LLaMA can all be used in summarization systems, but they have different architectures and capabilities. Therefore, choosing the correct model is important for a news organization.

1. BERT for News Summarization Architecture

BERT is a Transformer encoder-only model. It was originally designed mainly for language understanding rather than text generation.

BERT reads the words in a sentence using information from both directions. This allows it to understand the context of the article.

Working Principle

For summarization, BERT is commonly used for extractive summarization.

The article is divided into sentences. BERT creates contextual representations for the sentences and helps determine which sentences contain the most important information.

Example

Original Article:

"India's space agency successfully launched a new satellite on Monday. The satellite will provide improved weather forecasting services. Scientists said the mission will support climate monitoring."

Extractive Summary:

"India's space agency successfully launched a new satellite on Monday. The satellite will provide improved weather forecasting services."

The summary is created by selecting important sentences from the original article.

Advantages

1. Excellent contextual understanding.
2. Effective for identifying important sentences.
3. Good factual preservation because original sentences are selected.
4. Suitable for extractive summarization.
5. Can be fine-tuned for news-specific summarization.

Limitations

1. BERT is not naturally designed for generating new sentences.
2. Extractive summaries may contain unnecessary or repeated information.
3. It may not produce a smooth, human-like summary.
4. Long news articles can be challenging to process directly.
5. Usually requires task-specific fine-tuning.

Best Use

BERT is best suited for extractive summarization, especially when preserving the original wording and factual information is important.

2. GPT for News Summarization

Architecture

GPT uses a Transformer decoder-only architecture. It is primarily designed for generating text by predicting the next token based on the previous context.

Working Principle

The complete news article can be provided as input with an instruction such as:

"Summarize this news article in 3 sentences."

GPT analyzes the article and generates a new summary.

Example

Input:

"India's space agency successfully launched a new satellite on Monday. The satellite will improve weather forecasting and climate monitoring. Scientists stated that the mission will provide important environmental data."

GPT Summary:

"India successfully launched a new satellite designed to improve weather forecasting and climate monitoring. Scientists expect the satellite to provide valuable environmental data."

This is an abstractive summary because the model generates new wording.

Advantages

1. Strong language understanding and generation.
2. Can create fluent and readable summaries.
3. Can summarize articles using simple natural-language instructions.
4. Can produce summaries of different lengths and styles.
5. Can work with zero-shot or few-shot prompting.
6. Suitable for large-scale and diverse news content.

Limitations

1. May generate information that is not present in the article.
2. Hallucination can affect factual accuracy.
3. Large models can be computationally expensive.
4. Output can vary depending on the prompt.
5. Human or automated fact checking may be required for important news.

Best Use

GPT is suitable for flexible abstractive news summarization, particularly when summaries need to be fluent, concise and customizable.

3. T5 for News Summarization

Architecture

T5 stands for Text-to-Text Transfer Transformer. It uses an encoder-decoder Transformer architecture.

T5 treats different NLP problems as text-to-text problems. Summarization is naturally suited to this approach because the input is an article and the output is a shorter piece of text.

Working Principle

The article is provided with an instruction such as:

Input:

"Summarize: India successfully launched a new satellite..."

The encoder understands the article and the decoder generates the summary.

Example

Input:

"Summarize the following news article: India successfully launched a new satellite..."

Output:

"India successfully launched a satellite to improve weather forecasting and climate monitoring."

T5 can be fine-tuned specifically for summarization datasets.

Advantages

1. Designed for text-to-text tasks.
2. Naturally suitable for abstractive summarization.
3. Produces fluent and concise summaries.
4. Can be fine-tuned on news datasets.
5. Flexible for different NLP tasks.
6. Suitable for automated summarization pipelines.

Research and practical implementations commonly use T5 for abstractive summarization, and current Hugging Face documentation provides a complete T5 fine-tuning workflow for summarization.

Limitations

1. Fine-tuning may require significant computational resources.
2. May generate incorrect information.
3. Generated summaries require evaluation for factual accuracy.
4. Large versions can be expensive to run.
5. Very long news articles may require preprocessing or chunking.

Best Use

T5 is highly suitable for abstractive news summarization, especially when the organization wants a model that can be fine-tuned specifically for summarization.

4. LLaMA for News Summarization

Architecture

LLaMA is a family of decoder-only Transformer-based large language models. It is designed as a general-purpose language model and can be used for text generation and understanding tasks.

Working Principle

A news article can be provided together with an instruction:

"Generate a 3-sentence summary of this article and include only information present in the article."

LLaMA processes the article and generates an abstractive summary.

Example

Input:

"Summarize this news article in three sentences..."

Output:

"India successfully launched a new satellite on Monday. The satellite is expected to improve weather forecasting and climate monitoring. Scientists believe the mission will provide valuable environmental data."

Advantages

- 1. Strong text generation capabilities.
- 2. Can generate fluent and human-readable summaries.
- 3. Can be customized using fine-tuning.
- 4. Suitable for different summarization styles.
- 5. Can handle multiple NLP tasks using the same foundation model.
- 6. Open model versions provide opportunities for local deployment and customization.

Limitations

- 1. Large models require considerable computing resources.
- 2. May hallucinate information.
- 3. Summary quality depends on model size and prompting.
- 4. Long articles may require chunking or long-context processing.
- 5. Additional fact-checking may be necessary for news applications.

Best Use

LLaMA is suitable for customizable abstractive summarization, particularly when an organization wants more control over the model and deployment environment.

5. Comparison Table

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder-only	Decoder-only	Encoder-decoder	Decoder-only
Main Purpose	Language understanding	Text generation	Text-to-text tasks	General-purpose language generation
Summarization Type	Mainly extractive	Abstractive	Abstractive	Abstractive
Summary Fluency	Medium	Excellent	Very Good	Excellent
Factual Preservation	High for extractive summaries	Moderate	Good	Moderate–Good
Prompt-based Summarization	Limited	Excellent	Good	Excellent
Fine-tuning	Usually required	Optional	Useful	Optional/useful
Computational Cost	Low–Medium	High	Medium–High	High
Customization	Good	Good	Very Good	Excellent

Hallucination Risk	Low	Higher	Moderate	Higher
Best Strength	Important sentence selection	Flexible generation	Controlled text-to-text summarization	Customizable generation
Best Application	Extractive news summaries	Flexible AI news summaries	Fine-tuned news summarization	Custom/local AI summarization

6. Comparison of Working

BERT

News Article → Sentence Representation → Sentence Importance → Select Important Sentences → Extractive Summary

GPT

News Article + Prompt → Language Understanding → Text Generation → Abstractive Summary

T5

News Article + "Summarize" Task → Encoder → Decoder → Abstractive Summary

LLaMA

News Article + Instruction → Transformer Processing → Text Generation → Abstractive Summary

7. Real-World News Organization Scenario

Consider a news organization that receives 10,000 articles per day from different sources.

The organization wants to:

- Automatically summarize every article.
- Produce summaries in 2–5 sentences.
- Preserve important facts.
- Reduce reading time.
- Support different types of news.
- Process articles quickly.
- Provide summaries for a mobile application and website.

The four models can be considered as follows.

BERT

BERT can identify the most important sentences and create extractive summaries.

Advantage: Better preservation of original facts.

Problem: Summaries may not be as fluent or concise as generated summaries.

GPT

GPT can generate high-quality summaries from instructions.

Advantage: Excellent flexibility and readability.

Problem: It can sometimes introduce information that was not present in the original article.

T5

T5 can be fine-tuned specifically for news summarization.

Advantage: Good balance between understanding and generation.

Problem: Fine-tuning and deployment require computational resources.

LLaMA

LLaMA can generate summaries and can be customized for the organization's specific requirements.

Advantage: Strong generation and customization possibilities.

Problem: Large models require substantial computing resources and need safeguards against hallucination.

8. Evaluation Metrics

A news organization should not evaluate summarization quality only by reading a few examples. It should use automatic evaluation metrics and human evaluation.

ROUGE

ROUGE measures the overlap between the generated summary and reference summary.

Common variants include:

- ROUGE-1
- ROUGE-2
- ROUGE-L

BERTScore

BERTScore evaluates semantic similarity between the generated summary and reference summary.

Human Evaluation

Human reviewers can evaluate:

- Accuracy
- Relevance
- Fluency
- Coherence
- Factual consistency

Research on summarization commonly uses metrics such as ROUGE and BERTScore to compare models.

9. Strengths and Limitations Summary

Model	Strengths	Limitations
BERT	Accurate sentence selection, strong contextual understanding, good factual preservation	Mainly extractive, limited generation
GPT	Excellent fluency, flexible prompting, strong abstractive summaries	Hallucination, high computational/API cost
T5	Excellent text-to-text framework, fine-tunable for summarization	Requires resources for fine-tuning and inference
LLaMA	Strong generation, customizable, suitable for local deployment	High resource requirements, hallucination risk

10. Which Model Should the News Organization Choose?

The best choice depends on the organization's priorities.

If factual preservation is the highest priority:

BERT is a strong choice because extractive summarization selects sentences directly from the original article.

If maximum flexibility and natural language generation are required:

GPT is a strong choice because it can follow different summarization instructions and generate fluent summaries.

If the organization wants a dedicated, fine-tunable summarization system:

T5 is the most suitable choice.

If the organization wants an open and customizable large language model:

LLaMA is a strong choice.

11. Recommended Model

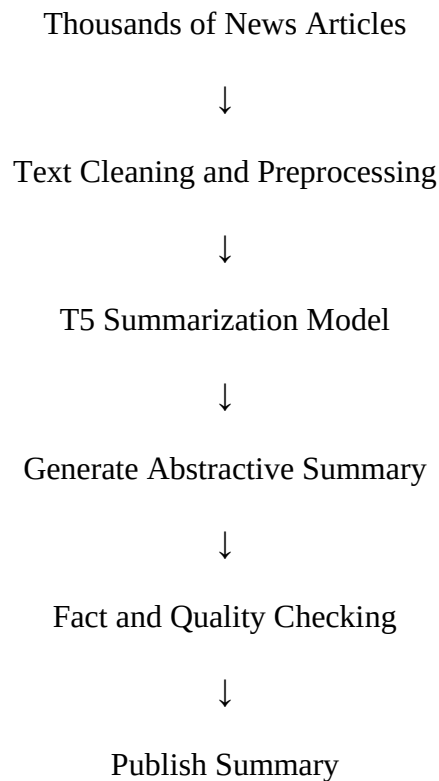
Recommended Model: T5

For the given scenario, T5 is the recommended model for automatically summarizing thousands of news articles.

The main reason is that T5's encoder-decoder architecture is naturally suited to converting a long article into a shorter generated summary. Its text-to-text framework allows the same model to be fine-tuned specifically for news summarization. Current NLP tooling also provides direct support for fine-tuning T5 for abstractive summarization.

A news organization can fine-tune T5 using a large collection of article-summary pairs. After training, the model can automatically generate summaries for new articles.

Recommended Workflow



This approach provides a good balance between summary quality, readability, customization and automation.

12. Alternative Hybrid Approach

A strong practical solution can also combine BERT and T5.

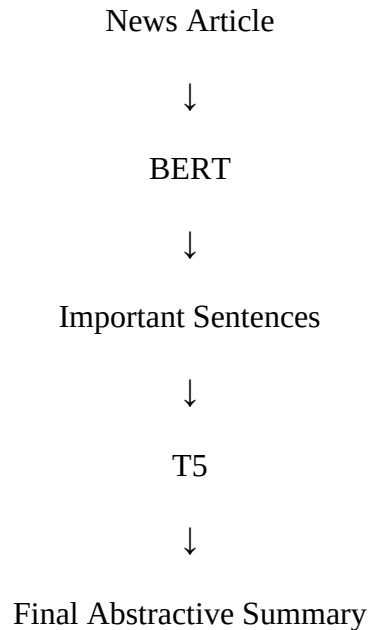
Step 1 – BERT

BERT identifies the most important sentences from the news article.

Step 2 – T5

T5 takes the selected information and generates a concise, fluent summary.

Workflow



A recent study has explored this type of BERT-based extractive + T5-based abstractive approach, reporting improvements in informativeness and coherence compared with individual approaches.

This hybrid approach can be particularly useful when factual preservation and readability are both important.

13. Final Conclusion

BERT, GPT, T5 and LLaMA are powerful Transformer-based models, but they have different strengths for news summarization.

BERT is effective for extractive summarization and is useful when factual preservation is the main priority.

GPT provides highly flexible and fluent abstractive summaries but may require additional safeguards against hallucination.

T5 is designed around a text-to-text framework and is highly suitable for fine-tuned abstractive summarization.

LLaMA provides powerful generative capabilities and is attractive when customization and deployment control are important.

For a news organization processing thousands of articles every day, T5 is recommended as the primary model because it provides a strong balance between contextual understanding, abstractive generation, fine-tuning capability and summarization performance.

For even stronger factual reliability, a BERT + T5 hybrid system can be considered, where BERT selects important information and T5 converts it into a concise and fluent final summary.

Therefore:

Best for extractive summarization → BERT

Best for flexible generative summarization → GPT

Best for fine-tuned news summarization → T5

Best for customizable open-model summarization → LLaMA

Recommended for this scenario → T5

References

- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT.
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RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Architectural Comparison Accuracy	30%	Accurate, in-depth comparison of architectures and use-cases	Accurate comparison with minor gaps	Superficial comparison of models	Inaccurate or confused comparison
Application Mapping	25%	Clearly maps each model to appropriate real-world applications	Reasonable mapping to applications	Weak mapping to applications	No meaningful mapping presented
Research Depth	20%	Well-researched with credible sources/benchmarks cited	Adequately researched	Limited research depth	Unresearched / opinion-based content
Delivery	15%	Confident, engaging, well-paced delivery	Clear delivery with minor pacing issues	Hesitant delivery	Unclear or unprepared delivery
Slide Design & References	10%	Well-designed slides with proper references	Adequate slides, minor reference gaps	Basic slides, poor referencing	No slides or references provided

Marks Evaluation:

Question No.	Architectural Comparison Accuracy (30%)	Application Mapping (25%)	Research Depth (20%)	Delivery (15%)	Slide Design & References (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						